Zhuocheng Zhang

zczhang1997@gmail.com | (+1) 765-761-6044 | Hong Kong | zhuochengzhang.com

Semiconductor Devices & Manufacturing | AI Agents for Semiconductor Engineering

Professional Summary

Semiconductor researcher and engineer with deep hands-on experience in device physics, fabrication, process integration, characterization and modeling. Cross-layer technical background spanning digital IC design, computer architecture, AI computing systems and LLM/agent engineering. Focused on applying AI to build agentic engineering workflows for semiconductor design, modeling and manufacturing.

Experience

Postdoctoral Researcher — The Chinese University of Hong Kong *2026 – Present*

AI agents for semiconductor compact modeling | Advisor: Prof. Bei Yu

- Built an agent-native compact-modeling system integrating SPICE tools, physics-aware guardrails, structured outputs and auditable execution traces for autonomous modeling workflows.
- Designed and executed a controlled evaluation on open model cards and measured silicon to quantify task completion, physics-consistent reasoning, and the effect of domain knowledge injection.
- Developed a governed knowledge layer and machine-readable ontology that captures evidence, modeling concepts, parameter relationships, QA rules and validated procedures for agent tasks.

Senior Device Engineer — HiSilicon, Huawei, China *2024 – 2025*

Vertical GAA silicon FET (VFET) development for advanced logic

- Designed VFET architectures using TCAD and COMSOL simulations, optimizing device geometry, doping and gate-stack parameters for 300-mm wafer fab tape-out.
- Drove device development through the full logic process flow with process-integration teams, defining DOE splits and reviewing inline characterization to evaluate process impact.
- Analyzed WAT data across lots and device geometries, validated root causes with targeted simulations, and translated findings into device and process changes for subsequent wafer runs.

Ph.D. Researcher — Purdue University, Birck Nanotechnology Center, USA *2019 – 2023*

Oxide-semiconductor transistors for monolithic 3D integration | Advisor: Prof. Peide Ye

- Investigated the material and device physics of ALD oxide semiconductors, studying thickness scaling, doping, annealing and reliability to understand carrier transport and guide transistor design.
- Fabricated planar, gate-all-around and vertical transistors with 5,000+ hours of hands-on cleanroom experience across lithography, etching, deposition, annealing and structural characterization.
- Characterized device performance through electrical, temperature-dependent and reliability measurements, linking experimental data to physical mechanisms and demonstrating GAA In_2O_3 FETs with near 20 mA/μm drain current.

Technical Skills

Semiconductor Devices & Manufacturing: device and semiconductor physics, GAA/VFET architectures, cleanroom fabrication and process integration, DOE, electrical characterization and WAT analysis, TCAD/COMSOL simulation, SPICE/BSIM compact modeling.

Digital IC Design & DTCO: systems-level understanding of architecture, RTL, verification, logic synthesis, STA, P&R, signoff, PDKs, standard cells, SRAM/DRAM, design enablement.

Computer Architecture & AI Systems: systems-level understanding of GPU architecture, CUDA execution and memory hierarchy, AI compiler stacks and kernel optimization, distributed training and LLM inference systems, PCIe/NVLink/RDMA and high-speed interconnects, operating-system memory and I/O.

AI/ML & Agent Engineering: machine learning and deep learning, Transformer/LLM architectures, SFT, PEFT, multimodal models, RAG, LLM agents, structured workflows and harness engineering.

Programming & Tools: Claude Code/Codex, Python, PyTorch, Hugging Face, LangGraph, MCP, FastAPI, Docker, Redis, Git, Milvus, MATLAB, Origin.

Technical Foundations: solid-state and semiconductor physics, reliability physics, electromagnetics, quantum mechanics, digital circuits, linear algebra, probability and statistics.

Education

Ph.D., Purdue University, USA *2019 – 2023*

Microelectronics, School of Electrical and Computer Engineering

B.S., University of Science and Technology of China, China *2014 – 2018*

Applied Physics, School of the Gifted Young

Selected Publications

Total citations: 1300+ | h-index: 21 | Full list: Google Scholar

Conference Proceedings

Z. Zhang, Z. Lin, C. Niu, M. Si, M. A. Alam and P. D. Ye, "Ultrahigh Bias Stability of ALD In_2O_3 FETs Enabled by High Temperature O_2 Annealing," in *IEEE Symposium on VLSI Technology and Circuits (VLSI)*, 2023.

Z. Zhang, Z. Lin, A. Charnas, H. Dou, Z. Shang, J. Zhang, M. Si, H. Wang, M. A. Alam and P. D. Ye, "Reliability of Atomic-Layer-Deposited Gate-All-Around In_2O_3 Nano-Ribbon Transistors with Ultra-High Drain Currents," in *IEEE International Electron Devices Meeting (IEDM)*, 2022.

Z. Lin, **Z. Zhang**, C. Niu, H. Dou, K. Xu, M. Islam, J.-Y. Lin, C. Sung, M. Hong, D. Ha, H. Wang, M. A. Alam and P. D. Ye, "Highly Robust All-Oxide Transistors with Ultrathin In_2O_3 as Channel and Thick In_2O_3 as Metal Gate Towards Vertical Logic and Memory," in *IEEE Symposium on VLSI Technology and Circuits (VLSI)*, 2024.

J. Zhang, **Z. Zhang**, H. Dou, Z. Lin, K. Xu, W. Yang, X. Zhang, H. Wang and P. D. Ye, "Fluorine Anion-Doped Ultra-Thin InGaO Transistors Overcoming Mobility-Stability Trade-off," in *IEEE International Electron Devices Meeting (IEDM)*, 2023.

J.-Y. Lin, **Z. Zhang**, S. Alajlouni, P.-Y. Liao, Z. Lin, C. Niu, A. Shakouri and P. D. Ye, "First Determination of Thermal Resistance and Thermal Capacitance of Atomic-Layer-Deposited In_2O_3 Transistors," in *IEEE International Electron Devices Meeting (IEDM)*, 2023.

Journal Articles

Z. Zhang, Z. Lin, P.-Y. Liao, V. Askarpour, H. Dou, Z. Shang, A. Charnas, M. Si, S. Alajlouni, A. Shakouri, H. Wang, M. Lundstrom, J. Maassen and P. D. Ye, "A Gate-All-Around In_2O_3 Nanoribbon FET with Near 20 mA/um Drain Current," *IEEE Electron Device Lett.,* vol. 43, no. 11, pp. 1905-1908, 2022.

Z. Zhang, Z. Lin, M. Si, D. Zhang, H. Dou, Z. Chen, A. Charnas, H. Wang and P. D. Ye, "Vertically Stacked Multilayer Atomic-layer-deposited Sub-1-nanometer In_2O_3 Field-effect Transistors with Back-end-of-line Compatibility," *Appl. Phys. Lett.*, vol. 120, no. 20, p. 202104, 2022.

Z. Zhang, Y. Hu, Z. Lin, M. Si, A. Charnas, K. Cho and P. D. Ye, "Atomically Thin Indium-Tin-Oxide Transistors Enabled by Atomic Layer Deposition," *IEEE Trans. Electron Devices*, vol. 69, no. 1, pp. 231-236, 2022.

J.-Y. Lin, **Z. Zhang**, Z. Lin, C. Niu, Y. Zhang, Y. Zhang, T. Kim, H. Jang, C. Sung, M. Hong, S. Lee, T. Lee, M. Cho, D. Ha, C. Jeong, H. Wang, M. A. Alam, and P. D. Ye, "First Demonstration of Top-Gate Enhancement-Mode ALD In_2O_3 FETs with High Thermal Budget of 600 °C for DRAM Applications," *IEEE Electron Device Lett.,* vol. 45, no. 10, pp. 1851-1854, 2024.

A. Charnas, **Z. Zhang**, Z. Lin, D. Zheng, J. Zhang, M. Si and P. D. Ye, "Extremely Thin Amorphous Indium Oxide Transistors," *Advanced Materials*, vol. 36, no. 9, p. 2304044, 2024.

C. Niu, **Z. Zhang**, D. Graf, S. Lee, M. Wang, W. Wu, T. Low and P. D. Ye, "High-pressure Induced Weyl Semimetal Phase in 2D Tellurium," *Communications Physics*, vol. 6, no. 1, p. 345, 2023.

M. Si, **Z. Zhang**, S.-C. Chang, N. Haratipour, D. Zheng, J. Li, U. E. Avci and P. D. Ye, "Asymmetric Metal/a-In_2Se_3/Si Crossbar Ferroelectric Semiconductor Junction," *ACS Nano*, vol. 15, no. 3, pp. 5689-5695, 2021.